

Accessibility and inclusion

Overdrive

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Summary

OverDrive – Accessibility and inclusion

1- Visual disability

2- Hearing disability

3- Motor disability

4- Cognitive disability

5. Specific constraints to virtual reality (VR / AR)

6- Mobile and web accessibility

7- Conclusion

The development of OverDrive integrates an in-depth reflection on accessibility from its design. The goal is to ensure that anyone, regardless of their physical, sensory or cognitive limitations, can access the essential features of the service.

Digital accessibility is not limited to compliance with technical standards: it is based on an inclusive approach aimed at reducing barriers to use and offering an experience adaptable to the diversity of users.

1- Handicap visuel

Identified constraints

- Difficulty reading complex or dense information
- Problems related to contrast or font size
- Potential incompatibilities with certain visual impairments (low vision, color blindness, eye strain)
- Difficulty identifying information conveyed solely through color

Measures envisaged

- High contrast between text and background
- Adjustable text size
- Simplified display mode that reduces secondary information
- Clear data hierarchy (visual prioritization)
- Avoid relying solely on color to convey information (e.g., flags, alerts, incidents)
- Mobile compatibility with screen readers

The VR interface favors clear readability, simple typography and well-defined display areas.

2- Hearing disability

Identified constraints

- Difficulty or inability to hear audible alerts
- Lack of access to any audio comments

Measures envisaged

- Visual indicators for important events (incident, overtaking, flag)
- Textual or symbolic notifications
- No mandatory reliance on sound
- Visual redundancy for all audio information

Thus, the experience remains fully functional even without sound activation.

3- Motor disability

Identified constraints

- Difficulty performing complex movements in AR/VR
- Limited range of motion
- Muscle fatigue
- Reduced precision in gesture interactions

Measures under consideration

- Simple interactions (pointer selection, click, or button)
- No need for complex or repetitive gestures
- Menus accessible via native headset controls
- Sufficient response time to avoid any time pressure
- Configurable interface to minimize unnecessary movement

The goal is to ensure that the user can interact with the interface without excessive effort.

4- Cognitive disability

Identified constraints

- Information overload
- Difficulty understanding technical data
- Need for a clear and simplified structure

Measures envisaged

- Simplified mode focused on essential information
- Clear data hierarchy (primary/secondary)
- Visual explanations of events (clear symbols, pictograms)
- AR interface customization: each user can manually add or remove widgets

Customization makes it possible to adapt the complexity of the interface to everyone's level of understanding.

5- Specific constraints on virtual reality

The immersive environment can generate

- Visual fatigue
- Motion sickness
- Spatial disorientation
- Sensory overload

Measures envisaged

- Limited fast animations
- Stable and fixed interface in the workspace
- Option to reduce visual effects
- Ability to disable certain dynamic elements
- No imposed session duration

Ergonomics favors stability and readability to limit discomfort.

6- Mobile and web accessibility

The mobile application and the web dashboard are important alternatives to VR.

Planned measures

- Responsive interface
- Compatibility with screen readers
- Adherence to accessibility best practices (contrast, clear navigation)
- Sufficiently large buttons and touch areas
- Structured and predictable navigation

These supports help ensure wider accessibility, including for users who do not have a VR headset.

Conclusion

OverDrive integrates a reflection on accessibility from the design phase in order to limit barriers related to disabilities:

- Visual
- Auditory
- Motor
- Cognitive

The objective goes beyond simple technical compliance: it is a matter of offering an inclusive, adaptable experience that respects the diversity of users.

This approach contributes to strengthening the ethical, responsible and sustainable dimension of the project, in line with its technological and societal ambition.